QPCode

Reg.No : .....
Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) MODEL QUESTION PAPER
FIFTH SEMESTER
DSE - MG5DSECFS307- CRITICAL INFRASTRUCTURE SECURITY AND FORENSICS
(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)
Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Very Short Answer Questions
Answer any **10** questions
Each question carries **2** marks

1.	Identify two functions of SCADA systems in monitoring critical infrastructure operations.	[Understand (U)] / [CO1]
2.	Explain the relationship between Critical Infrastructure Protection and Cyber Crime.	[Understand (U)] / [CO1]
3.	Differentiate between critical infrastructure vulnerabilities and Internet infrastructure vulnerabilities.	[Understand (U)] / [CO1]
4.	Identify two common cyber vulnerabilities in critical infrastructure systems.	[Understand (U)] / [CO1]
5.	Differentiate between natural threats and intentional threats to critical infrastructure with one example each	[Analyse (An)] / [CO2]
6.	Compare the use of a risk matrix with probability modeling	[Analyse (An)] / [CO2]
7.	Differentiate between subjective judgment and objective data in risk analysis.	[Analyse (An)] / [CO2]
8.	Outline two key security objectives for critical infrastructure	[Analyse (An)] / [CO2]
9.	Inspect the role of law enforcement agencies in coordinating with private sector operators during infrastructure incidents.	[Analyse (An)] / [CO3]
10.	Differentiate between proactive preparedness measures and reactive response strategies in national security.	[Analyse (An)] / [CO3]
11.	Evaluate the effectiveness of redundancy (backup systems) as a mitigation strategy for physical threats.	[Evaluate(E)] / [CO4]
12.	Explain why physical security is considered a foundational element in protecting critical assets.	[Understand (U)] / [CO4]

[2x10 = 20]

Part B
Short Answer Questions
Answer any **6** questions
Each question carries **5** marks

13.	Explain the major types of attacks that can affect Internet Infrastructure.	[Understand (U)] / [CO1]
14.	Explain the scientific and technological nature of critical infrastructure	[Understand (U)] / [CO1]
15.	Summarize the importance of Critical Infrastructure Protection in protecting essential services from cyber threats.	[Understand (U)] / [CO1]
16.	Examine how Monte Carlo simulations enhance quantitative risk analysis.	[Analyse (An)] / [CO2]
17.	Predict how emerging technologies (IoT, AI, smart grids) may reshape future infrastructure development	[Analyse (An)] / [CO3]
18.	Categorize terrorist threats into biological, chemical, radiological, cyber, and physical domains, giving one example for each.	[Analyse (An)] / [CO3]
19.	A critical infrastructure facility has identified several potential physical threats. Assess the factors that should be considered when conducting a threat assessment	[Evaluate(E)] / [CO4]
20.	Rank different prevention measures (surveillance, barriers, access control) in terms of their effectiveness, and support your ranking with examples	[Evaluate(E)] / [CO4]

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **10** marks

21.	Examine how the risk management framework can improve security and resilience in critical infrastructure.	[Analyse (An)] / [CO2]
22.	Inspect the role of law enforcement agencies in detecting and preventing cyber and physical threats to infrastructure.	[Analyse (An)] / [CO3]
23.	Evaluate the potential impact of physical threats such as unauthorized access, vandalism, and deliberate attacks on critical assets.	[Evaluate (E)] / [CO4]

[10x2 = 20]